



10165 - Blue oases in a hostile red desert

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
SECCO1				
	1	SECCO1-NIRCam	NIRCam Imaging	(1) SECCO1

Folder	Observation	Label	Observing Template	Science Target
	2	SECCO1-MIRI	MIRI Imaging	(1) SECCO1
BC3				
	3	BC3-NIRCam	NIRCam Imaging	(3) BC3
	4	BC3-MIRI	MIRI Imaging	(3) BC3
BC4				
	5	BC4-NIRCam	NIRCam Imaging	(4) BC4
	6	BC4-MIRI	MIRI Imaging	(4) BC4
BC1				
	7	BC1-NIRCam	NIRCam Imaging	(2) BC1
	8	BC1-MIRI	MIRI Imaging	(2) BC1
BC5				
	9	BC5-NIRCam	NIRCam Imaging	(5) BC5
	10	BC5-MIRI	MIRI Imaging	(5) BC5

ABSTRACT

We request 26 h of NIRCam and MIRI time to image five recently discovered novel stellar systems in the Virgo galaxy cluster. These young, star-forming, metal-rich systems contain very few stars ($M^* < 10^5 M_{\text{sol}}$) and are remarkably isolated, like tiny blue oases in the enormous red desert of the Virgo cluster. They are likely the ram pressure stripping analogs of tidal dwarf galaxies, forming from stripped gas and somehow traversing 100s of kpc through the hot intra-cluster medium, while retaining enough cool gas to continue forming stars. They may also be the first examples of star formation (SF) in isolated cloudlets supported by external pressure. Existing HST observations of these systems found only young stars (< 100 Myr), but are not deep enough to rule out an older stellar population. These limits could not be improved without a prohibitive amount of HST time. With NIRCam we will reach a depth of 1 mag below the tip of the red giant branch and thus be able to conclusively identify or rule out the existence of any older population. Via fitting of the NIRCam CMD, we will also measure the SF history of each target over the past 500 Myr, revealing when stars first began forming, and providing crucial information for hydrodynamical simulations studying their formation mechanisms. Finally, MIRI imaging will map hot dust and embedded SF, which, in comparison to existing H-alpha maps and stars selected from the NIRCam CMD, will reveal how SF has proceeded. This novel class of objects provides a new window into the gas stripping processes that impact all gas-bearing galaxies in a cluster environment, but only with JWST can we understand the details of how they form.

OBSERVING DESCRIPTION

JWST Proposal 10165 (Created: Wednesday, June 3, 2026, 6:00:11PM Eastern Standard Time) - Overview

The proposed observations are NIRCам (F090W, F150W, and F277W) and MIRI (F770W) imaging of five enigmatic, young, blue, isolated, and faint stellar systems embedded in the hot intra-cluster medium of the Virgo galaxy cluster at a distance of 16.5 Mpc. These systems all have rich metallicities, indicating that they formed from stripped gas that was pre-enriched in a galaxy. However, they have apparently survived traversing the intra-cluster medium long enough to become remarkably isolated (>100 kpc), and were likely held together in part by external pressure support.

Existing HST imaging implies that the stellar populations of these systems are comprised entirely of young (<100 Myr) stars. However, older stellar populations cannot be ruled out with the depth afforded by HST. Deep NIRCам imaging (F090W=28 mag at 5-sigma) will reveal stars down to 1 mag below the theoretical tip of the red giant branch (and to the main sequence turn off for the youngest stars, <30 Myr), and will therefore be capable of ruling out (or revealing) the existence of any old (>1 Gyr) stellar population. These observations will thus be able to conclusively demonstrate that these are purely young stellar systems that must have formed from stripped gas. Fitting of the NIRCам CMDs will also provide a star formation history over the past 500 Myr, indicating when star formation was ignited and providing vital clues as to the formation process of these objects.

Each target will be placed in module B of NIRCам and imaged over a 7-pt sub-pixel dither, with 7 groups per integration, and the MEDIUM8 read-out mode in both F090W and F150W. In both cases the LW filter will be F277W. Module A will be used to characterize field stars. PA constraints are needed for targets BC4 and BC5 to ensure that none of their components fall in the sub-module gaps. Additionally, SECCO1 and BC3 have PA constraints, such that dwarf galaxies that are projected nearby can be included in the observations.

The MIRI observations will reach a depth of 1 uJy at F770W (at 5-sigma), and will provide a map of the youngest embedded star formation regions in each target. In combination with existing H-alpha maps and the location of stars selected from the NIRCам CMDs this will reveal how star formation has proceeded in these objects.

For the MIRI observations we will use the FASTR1 read-out mode with a 4-pt dither pattern over a total exposure time of ~ 25 min per target. The targets are placed to one side of the MIRI FoV so that the remainder can be used for sky characterization. For BC4 and BC5 this requires PA constraints to ensure that all components are included.

Proposal 10165 - Targets - Blue oases in a hostile red desert

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SECCO1	RA: 12 21 53.9000 (185.4745833d) Dec: +13 27 37.00 (13.46028d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Unidentified</i> <i>Description=[Radio sources, Ultraviolet sources, Visible sources]</i>		
	(2)	BC1	RA: 12 39 2.0000 (189.7583333d) Dec: +12 12 16.70 (12.20464d) Equinox: J2000 <i>Comments:</i> <i>Category=Unidentified</i> <i>Description=[Ultraviolet sources, Visible sources]</i>		
	(3)	BC3	RA: 12 46 42.6395 (191.6776646d) Dec: +10 22 7.47 (10.36874d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Unidentified</i> <i>Description=[Radio sources, Ultraviolet sources, Visible sources]</i>		
	(4)	BC4	RA: 12 26 23.7729 (186.5990537d) Dec: +14 23 21.60 (14.38933d) Equinox: J2000 <i>Comments:</i> <i>Category=Unidentified</i> <i>Description=[Ultraviolet sources, Visible sources]</i>		
	(5)	BC5	RA: 12 26 30.9000 (186.6287500d) Dec: +15 10 26.20 (15.17394d) Equinox: J2000 <i>Comments:</i> <i>Category=Unidentified</i> <i>Description=[Ultraviolet sources, Visible sources]</i>		

Proposal 10165 - Observation 1 - Blue oases in a hostile red desert

Observation	Proposal 10165, Observation 1: SECCO1-NIRCam									Wed Jun 03 23:00:12 GMT 2026
	Diagnostic Status: Warning									
Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(1)	SECCO1	RA: 12 21 53.9000 (185.4745833d) Dec: +13 27 37.00 (13.46028d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Unidentified</i> <i>Description=[Radio sources, Ultraviolet sources, Visible sources]</i>							
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module B center (small extended source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				STANDARD				7
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F090W	F277W	MEDIUM8	7	1	7	7	5110.703	274401
	2	F150W	F277W	MEDIUM8	7	1	7	7	5110.703	274401
Special Requirements	Aperture PA Range 106.05262691 to 110.05262691 Degrees (V3 106.0 to 110.0) Offset -29.48250616789052 arcsec, 55.67518874156603 arcsec Fiducial Point Override NRCBS_FULL									

Proposal 10165 - Observation 2 - Blue oases in a hostile red desert

Observation	Proposal 10165, Observation 2: SECCO1-MIRI										Wed Jun 03 23:00:12 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SECCO1	RA: 12 21 53.9000 (185.4745833d) Dec: +13 27 37.00 (13.46028d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Unidentified</i> <i>Description=[Radio sources, Ultraviolet sources, Visible sources]</i>								
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	5		1	1			DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	110	1	1	Dither 1	5	5	1526.272	274402

Proposal 10165 - Observation 3 - Blue oases in a hostile red desert

Observation	Proposal 10165, Observation 3: BC3-NIRCam									Wed Jun 03 23:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(3)	BC3	RA: 12 46 42.6395 (191.6776646d) Dec: +10 22 7.47 (10.36874d) Equinox: J2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Unidentified Description=[Radio sources, Ultraviolet sources, Visible sources]									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module B center (small extended source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				STANDARD				7
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F090W	F277W	MEDIUM8	7	1	7	7	5110.703	274401
	2	F150W	F277W	MEDIUM8	7	1	7	7	5110.703	274401
Special Requirements	Aperture PA Range 65.05262691 to 180.05262691 Degrees (V3 65.0 to 180.0) Offset -27.31163961227381 arcsec, 38.13266725849424 arcsec Fiducial Point Override NRCBS_FULL									

Proposal 10165 - Observation 4 - Blue oases in a hostile red desert

Observation	Proposal 10165, Observation 4: BC3-MIRI Diagnostic Status: Warning Observing Template: MIRI Imaging										Wed Jun 03 23:00:12 GMT 2026
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	BC3	RA: 12 46 42.6395 (191.6776646d) Dec: +10 22 7.47 (10.36874d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Unidentified</i> <i>Description=[Radio sources, Ultraviolet sources, Visible sources]</i>								
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	5		1	1			DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	110	1	1	Dither 1	5	5	1526.272	274402
Special Requirements	Offset -2.401460125441556 arcsec, -0.18225502814532155 arcsec										

Proposal 10165 - Observation 5 - Blue oases in a hostile red desert

Observation	Proposal 10165, Observation 5: BC4-NIRCam									Wed Jun 03 23:00:12 GMT 2026	
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(4)	BC4	RA: 12 26 23.7729 (186.5990537d) Dec: +14 23 21.60 (14.38933d) Equinox: J2000								
	Comments: Category=Unidentified Description=[Ultraviolet sources, Visible sources]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module B center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				7	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F277W	MEDIUM8	7	1	7	7	5110.703	274401	
	2	F150W	F277W	MEDIUM8	7	1	7	7	5110.703	274401	
Special Requirements	Aperture PA Range 85.05262691 to 112.05262691 Degrees (V3 85.0 to 112.0) Aperture PA Range 133.05262691 to 143.05262691 Degrees (V3 133.0 to 143.0) Offset 31.902735052474103 arcsec, 6.0874079076044865 arcsec Fiducial Point Override NRCBS_FULL										

Proposal 10165 - Observation 6 - Blue oases in a hostile red desert

Observation	Proposal 10165, Observation 6: BC4-MIRI										Wed Jun 03 23:00:12 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous		
	(4)	BC4	RA: 12 26 23.7729 (186.5990537d) Dec: +14 23 21.60 (14.38933d) Equinox: J2000								
	Comments: Category=Unidentified Description=[Ultraviolet sources, Visible sources]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets	1	5		1	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	135	1	1	Dither 1	4	4	1498.522	274402
Special Requirements	Aperture PA Range 81.83544897 to 147.83544897 Degrees (V3 77.0 to 143.0) Offset -4.367589537302167 arcsec, 19.45148289086866 arcsec										

Proposal 10165 - Observation 7 - Blue oases in a hostile red desert

Observation	Proposal 10165, Observation 7: BC1-NIRCam										Wed Jun 03 23:00:12 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	BC1	RA: 12 39 2.0000 (189.7583333d) Dec: +12 12 16.70 (12.20464d) Equinox: J2000								
	Comments: Category=Unidentified Description=[Ultraviolet sources, Visible sources]										
Template	Module			Subarray			Target Placement				
	ALL			FULL			Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				7	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F277W	MEDIUM8	7	1	7	7	5110.703	274401	
	2	F150W	F277W	MEDIUM8	7	1	7	7	5110.703	274401	
Special Requirements	Aperture PA Range 176.34481904 to 141.34481904 Degrees (V3 176.0 to 141.0) Fiducial Point Override NRCB1_FULL										

Proposal 10165 - Observation 8 - Blue oases in a hostile red desert

Observation	Proposal 10165, Observation 8: BC1-MIRI										Wed Jun 03 23:00:12 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous		
	(2)	BC1	RA: 12 39 2.0000 (189.7583333d) Dec: +12 12 16.70 (12.20464d) Equinox: J2000								
Template	Comments: Category=Unidentified Description=[Ultraviolet sources, Visible sources]										
	Subarray FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	5		1	1			DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	110	1	1	Dither 1	5	5	1526.272	274402
Special Requirements	Offset -2.7191504420586257 arcsec, 1.4990053597432882 arcsec										

Proposal 10165 - Observation 9 - Blue oases in a hostile red desert

Observation	Proposal 10165, Observation 9: BC5-NIRCam									Wed Jun 03 23:00:12 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(5)	BC5	RA: 12 26 30.9000 (186.6287500d) Dec: +15 10 26.20 (15.17394d) Equinox: J2000							
	Comments: Category=Unidentified Description=[Ultraviolet sources, Visible sources]									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module B center (small extended source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				STANDARD				7
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F090W	F277W	MEDIUM8	7	1	7	7	5110.703	274401
	2	F150W	F277W	MEDIUM8	7	1	7	7	5110.703	274401
Special Requirements	Aperture PA Range 27 to 73 Degrees (V3 26.94737309 to 72.94737309) Aperture PA Range 118.05262691 to 160.05262691 Degrees (V3 118.0 to 160.0) Aperture PA Range 189.05262691 to 193.0 Degrees (V3 189.0 to 192.94737309) Aperture PA Range 265 to 282 Degrees (V3 264.94737309 to 281.94737309) Aperture PA Range 358 to 4 Degrees (V3 357.94737309 to 3.94737309) Offset 36.55964375572251 arcsec, -27.28275007318709 arcsec Fiducial Point Override NRCBS_FULL									

Proposal 10165 - Observation 10 - Blue oases in a hostile red desert

Observation	Proposal 10165, Observation 10: BC5-MIRI										Wed Jun 03 23:00:12 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous		
	(5)	BC5	RA: 12 26 30.9000 (186.6287500d) Dec: +15 10 26.20 (15.17394d) Equinox: J2000								
	Comments: Category=Unidentified Description=[Ultraviolet sources, Visible sources]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	5		1	1			DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	110	1	1	Dither 1	5	5	1526.272	274402
Special Requirements	Aperture PA Range 112.83544897 to 164.83544897 Degrees (V3 108.0 to 160.0) Offset 1.2049672766443256 arcsec, -5.5054208271032286 arcsec										